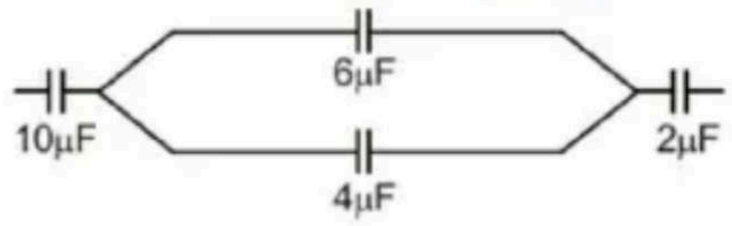
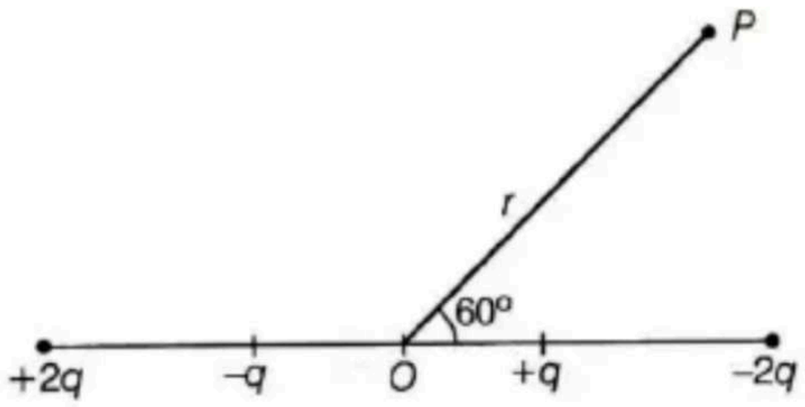
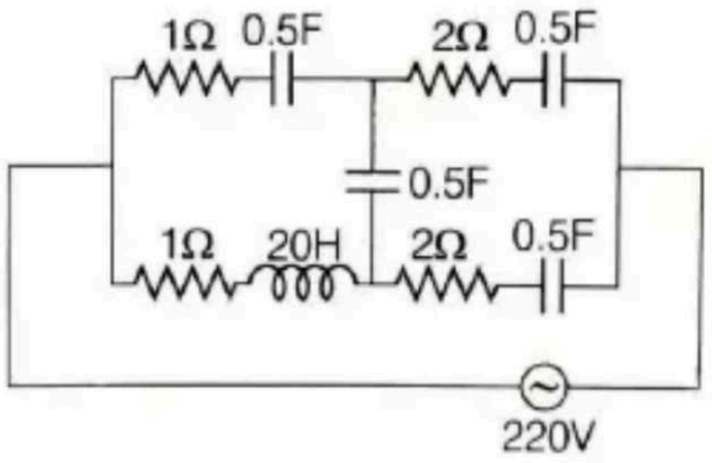
Infinity Learn by Sri Chaitanya Educational Institutions

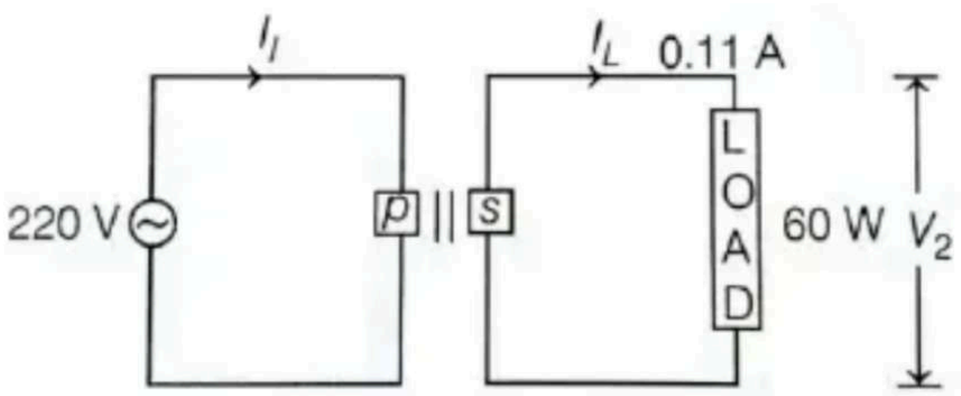
THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

Sri Chaitanya Educational Institutions | Infinity Learn | 38


6μF
10μF
2μF
4μF


P
r
60°
+2q
−q
O
+q
−2q


1Ω
0.5F
2Ω
0.5F
0.5F
1Ω
20H
2Ω
0.5F
220V


I_1
I_L
0.11 A
220 V
P
S
LOAD
60 W
V_2

—

—

—

—

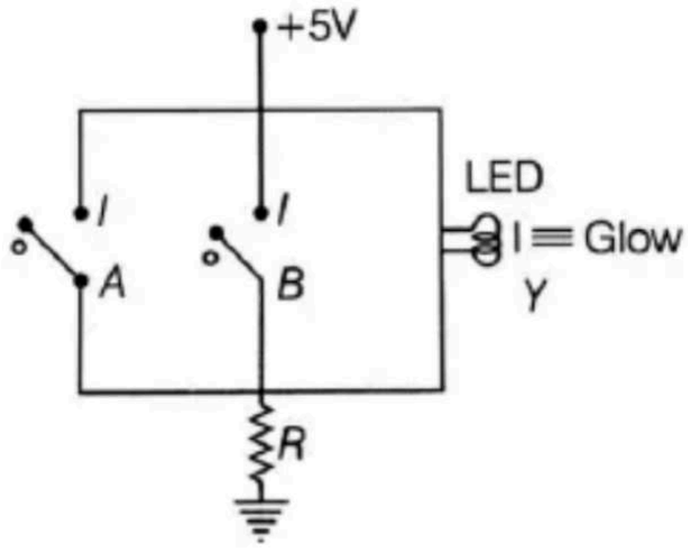
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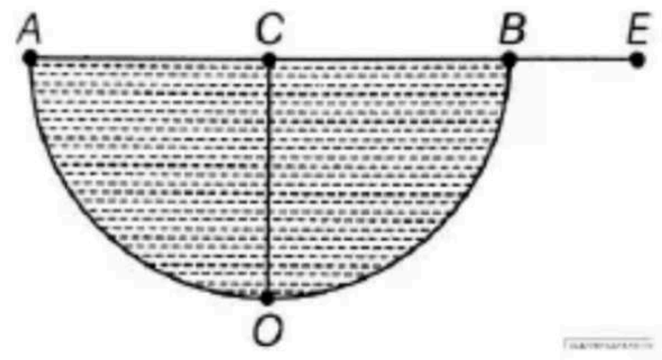
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+5V
+5V
D_1
D_2
V_{out}


+5V
I
I
A
B
LED
I ≡ Glow
Y
R

	—		
	—		
	—		

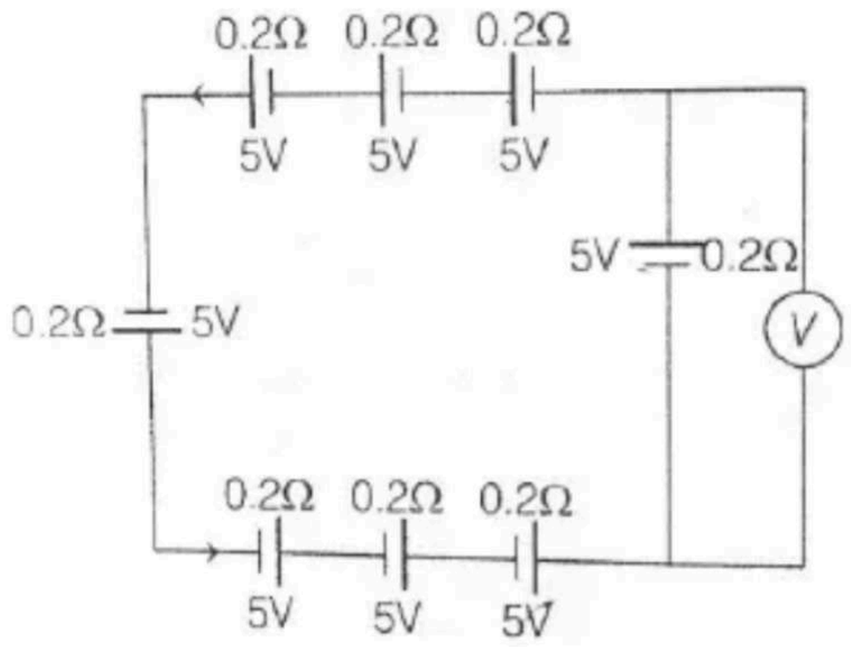

A
C
B
E
O

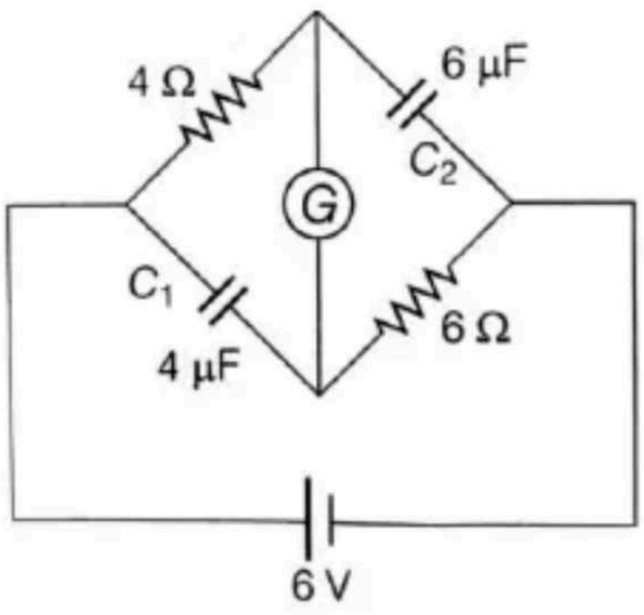
√

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√

√


0.2Ω
0.2Ω
0.2Ω
5V
5V
5V
5V
0.2Ω
0.2Ω
5V
V
0.2Ω
0.2Ω
0.2Ω
5V
5V
5V


4 Ω
6 μF
C_2
G
C_1
4 μF
6 Ω
6 V

24. Total charge 'Q' spread over the volume of sphere uniformly. Then flux through Gaussian surface which encloses a cube, cut from this sphere such that it has maximum possible volume is

(1) $\frac{3Q}{2\pi\varepsilon_0}$ (2) $\frac{2Q}{3\pi\varepsilon_0}$

(3) $\frac{\sqrt{3}Q}{2\pi\varepsilon_0}$ (4) $\frac{2Q}{\sqrt{3}\pi\varepsilon_0}$

25. A heavy ball of mass M is suspended from the ceiling of a car by a light string of mass m(m<<M). When the car is at rest, the speed of transverse waves in the string is $3ms^{-1}$. When the car has acceleration a, the wave speed increases to $4ms^{-1}$. The value of a is n times that of gravitational acceleration g then n is

(1) $\frac{7\sqrt{5}}{9}$ (2) $\frac{5\sqrt{7}}{9}$

(3) $\frac{7}{9\sqrt{5}}$ (4) $\frac{9}{5\sqrt{7}}$

26. Assuming the earth to be a sphere of uniform mass density, a body weighed 300 N on the surface of earth. The weight of the body at R/4 depth under surface of earth is

(1) 225 kg wt (2) 22.5 kg wt

(3) 200 kg wt (4) 20 kg wt

27. The minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of 2R is

(1) $\frac{5GmM}{6R}$ (2) $\frac{2GmM}{3R}$

(3) $\frac{GmM}{2R}$ (4) $\frac{GmM}{3R}$

28. The displacement of simple harmonic oscillator after 3 s starting from its mean position is equal to half of its amplitude. The time period of harmonic motion is

(1) 6 s (2) 8 s

(3) 12 s (4) 36 s

29. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion(A): Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason(R): Because the angle of contact between the oil/ grease and water is obtuse.

(1) Both A and R are true and R is the correct explanation of (A).

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

30. The terminal velocity v_t of the spherical rain drop depends on the radius r as

(1) $r^{1/2}$ (2) r

(3) r^2 (4) r^3

31. Two wires of same length and radius are joined end-to-end and loaded. The Young's moduli of the materials of the two wires are Y_1 and Y_2. The combination behaves as a single wire, then its Young's modulus is

(1) $Y = \frac{2Y_1Y_2}{3\ Y_1 + Y_2}$ (2) $Y = \frac{2Y_1Y_2}{Y_1 + Y_2}$

(3) $Y = \frac{Y_1Y_2}{2\ Y_1 + Y_2}$ (4) $Y = \frac{Y_1Y_2}{Y_1 + Y_2}$

32. During the melting of a slab of ice at 273 K at atmospheric pressure.

(1) Internal energy of the ice-water system decreases.

(2) Positive work is done by the ice-water system on the atmosphere.

(3) Positive work is done on the ice-water system by the atmosphere.

(4) Internal energy of ice-water system remains unchanged.

33. The correct relation between $\gamma = \frac{C_P}{C_V}$ and temperature Tis

(1) $\gamma \propto T$ (2) $\gamma \propto \frac{1}{T}$

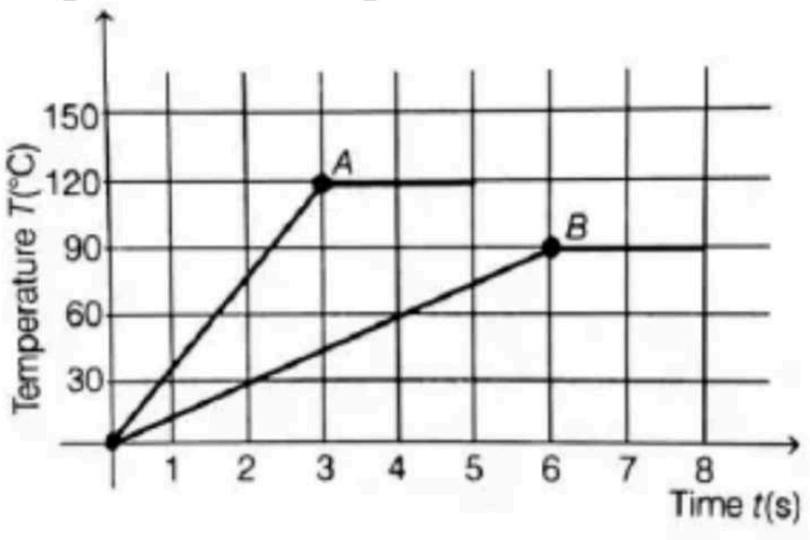
(3) $\gamma \propto T^0$ (4) $\gamma \propto \frac{1}{\sqrt{T}}$

34. The temperature of a body in air falls from 40°C to 24°C in 4 min. The temperature of the air is 16°C. The temperature of the body in the next 4 min will be

(1) $\frac{28}{3}$ °C (2) $\frac{42}{3}$ °C

(3) $\frac{14}{3}$ °C (4) $\frac{56}{3}$ °C

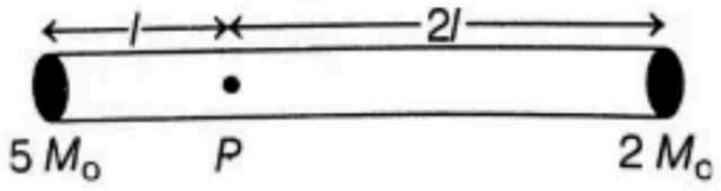
35. Two different metal bodies A and B of equal mass are heated at a uniform rate under similar conditions. The variation of temperature of the bodies is graphically represented as shown in the figure. The ratio of specific heat capacities of A & B is



(1) $\frac{8}{3}$ (2) $\frac{3}{8}$

(3) $\frac{3}{4}$ (4) $\frac{4}{3}$

36. A rigid massless rod of length $3l$ has two masses attached at each end as shown in the figure. The rod is pivoted at point P on the horizontal axis (see figure). When released from initial horizontal position, its instantaneous angular acceleration will be



(1) g / 13l (2) g / 2l

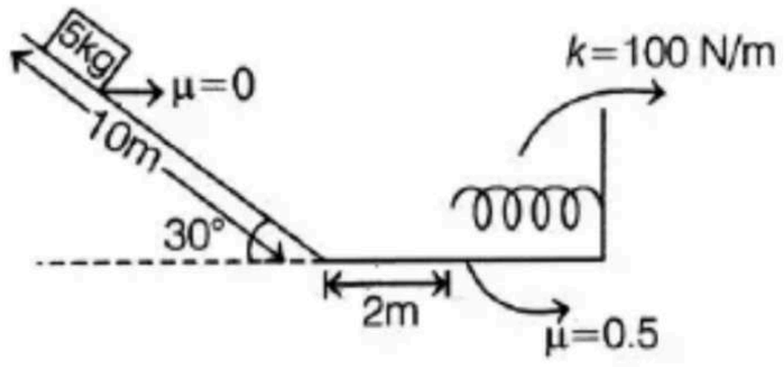
(3) 7g / 3l (4) g / 3l

37. Consider a two particle system with particles having masses m_1 and m_2. If the first particle is pushed towards the centre of mass through a distance d, the distance through which the second particle is to be moved, so as to keep the centre of mass at the same position is

(1) $\frac{m_2}{m_1}d$ (2) $\frac{m_1}{m_1+m_2}$

(3) $\frac{m_1}{m_2}d$ (4) d

38. A block is simply released from the top of an inclined plane, slides on to rough horizontal surface as shown in the figure below. The maximum compression in the spring when the block hits the spring is nearer to



(1) 1.45 m (2) 1.25 m

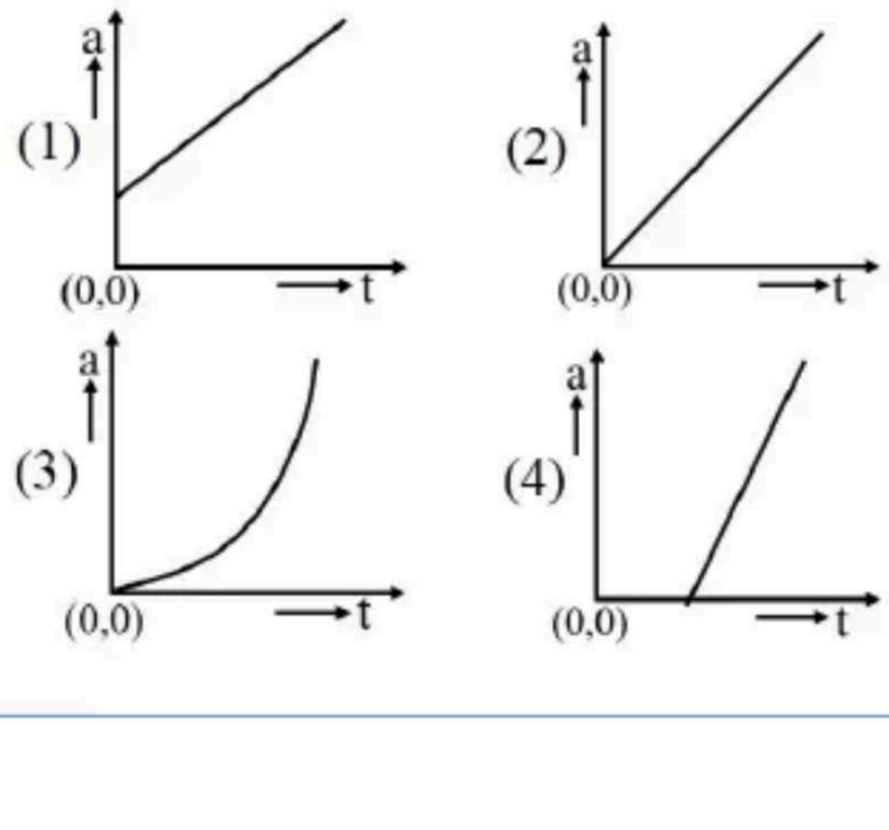
(3) 1.55 m (4) 1.75 m

39. A body of mass m connected to a massless and unstretchable string goes in vertical circle of radius R under gravity g. The other end of the string is fixed at the centre of circle. If velocity at top of circular path is $n\sqrt{gR}$ where, $n \geq 1$, then ratio of kinetic energy of the body at bottom to that at top of the circle is

(1) $\frac{n+4}{n}$ (2) $\frac{n^2}{n^2+4}$

(3) $\frac{n}{n+4}$ (4) $\frac{n^2+4}{n^2}$

40. A wooden block, initially at rest on the rough horizontal ground, is pushed by a force which increases linearly with time t. Which of the following curve best describes acceleration of the block with time?



41. A body of mass M at rest explodes into three pieces, in the ratio of masses 1:1:2. Two smaller pieces fly off perpendicular to each other with velocities of $30ms^{-1}$ and $40ms^{-1}$ respectively. The velocity of the third piece will be

(1) $15\ ms^{-1}$ (2) $25\ ms^{-1}$

(3) $35\ ms^{-1}$ (4) $50\ ms^{-1}$

42. A projectile is launched at an angle α with the horizontal with a velocity $20ms^{-1}$. After 10s, its inclination with horizontal is β. The value of tan β will be ($g = 10m^{-2}$)

(1) tan α + 5sec α (2) tan α – 5sec α

(3) 2tan α – 5sec α (4) 2tan α + 5sec α

43. A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10m in *t* s. the distance travelled by the toy in the next *t* s will be

(1) 10 m (2) 20 m

(3) 30 m (4) 40 m

44. If μ_0 and ε_0 are the permeability and permittivity of free space, respectively, then the dimensions of $\left(\frac{1}{\mu_0\varepsilon_0}\right)$ is

(1) $\left[\frac{L^2}{T^2}\right]$ (2) $\left[\frac{L}{T}\right]$

(3) $\left[\frac{T^2}{L^2}\right]$ (4) $\left[\frac{T^2}{L}\right]$

45. If the screw on a screw gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale, the least count of the screw gauge is

(1) 0.1 cm (2) 0.01 cm

(3) 0.02 cm (4) 0.001 cm

CHEMISTRY

46. Radius of first orbit in hydrogen atom is a_0. Radius of third orbit in Li^{2+} ion is

(1) $3a_0$ (2) a_0

(3) $\frac{a_0}{6}$ (4) $6a_0$

47. Identify correct statements.

A) Size of $Fe > Fe^{2+} > Fe^{3+}$

B) Ionization enthalpy of $Be > B > Li$

C) Electronegativity of $O > Cl > Br$

D) Acidic strength of $SO_3 > P_2O_5 > SiO_2$

(1) A, B and C only (2) A,B,C and D

(3) B and C only (4) B,C and D only

48. He_2, Be_2, B_2, C_2. How many species can exist

(1) zero (2) 2

(3) 3 (4) 1

49. Two moles of Nitric oxide reacts with 6.023×10^{23} molecules of dioxygen to produce _____grams of Nitrogen dioxide.

(1) 115 g (2) 46 g

(3) 69 g (4) 92 g

50. All the following are disproportionation reactions except

(1) $H_2S + SO_2 \longrightarrow S + H_2O$

(2) $Cu^{+} \longrightarrow Cu^{2+} + Cu$

(3) $S_8 \xrightarrow{OH^-} S^{2-} + S_2O_3^{2-}$

(4) $MnO_4^{2-} \xrightarrow{H^+} MnO_4^{-} + MnO_2$

51. Assertion (A) :

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, formation of NH_3 is favoured at high pressure (according to Lechatelier's principle)

Reason(R): $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$,

Formation of NH_3 is an exothermic reaction.

(1) A and R are true, R explains A

(2) A and R are true, R does not explain A

(3) A is true but R is false

(4) A is false but R is true

52. Dissociation constant of 1% ionized 0.2 M NH_4OH is

(1) 4×10^{-4} (2) 2×10^{-2}

(3) 2×10^{-5} (4) 4×10^{-6}

53. $N_2 + 3H_2 \rightarrow 2NH_3 \quad \Delta H = -93$ kJ/mole

$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O \quad \Delta H = -904$ kJ/mole

$N_2 + O_2 \rightarrow 2NO \quad \Delta H = +189$ kJ/mole

Enthalpy of formation of water is

(1) −142.36 kJ/mole (2) 118.39 kJ/mole

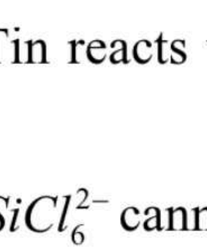
(3) −543.26 kJ/mole (4) −244.26 kJ/mole

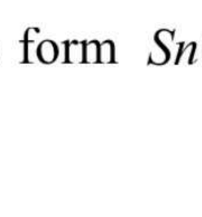
54. Assertion (A) : Gallium is used in high temperature thermometers.
Reason (R) : Gallium has wide liquid range
(1) A and R are true, R explains A
(2) A and R are true, R does not explain A
(3) A is true but R is false
(4) A is false but R is true

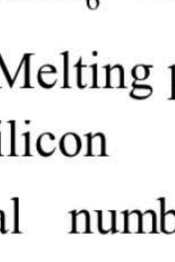
55. Incorrect statement among the following is
(1) CO is a neutral oxide
(2) Tin reacts with steam to form SnO_2 and H_2
(3) $SiCl_6^{2-}$ cannot exist
(4) Melting point of carbon is lesser than that of silicon

56. Total number of isomeric cyclic ethers possible with the molecular formula, C_3H_6O is
(1) 1 (2) zero
(3) 3 (4) 4

57.

A) 

B) 

C) 

Correct order of stability is
(1) C > A > B (2) B > C > A
(3) B > A > C (4) C > B > A

58. 2,3 Dimethyl but-2-ene

$\xrightarrow[\Delta]{KMnO_4/H^+} A \xrightarrow[H_2O]{NaBH_4} B$. 'B' can be prepared by

(1) $CH_3CHO \xrightarrow[H_2O]{CH_3MgBr}$

(2) $HCHO \xrightarrow[H_2O]{CH_3CH_2MgBr}$

(3) $CH_3-CH=CH_2 \xrightarrow[II)H_2O_2/OH^-]{I)B_2H_6/THF}$

(4) $CH_3COOC_2H_5 \xrightarrow{H_2/catalyst}$

59. Assertion (A) : Toluene is more reactive than Benzene towards electrophilic substitution
Reason (R) : $-CH_3$ group attached to benzene ring shows hyper conjugation and positive Inductive effect on benzene ring
(1) A and R are true, R explains A
(2) A and R are true, R does not explain A
(3) A is true but R is false
(4) A is false but R is true

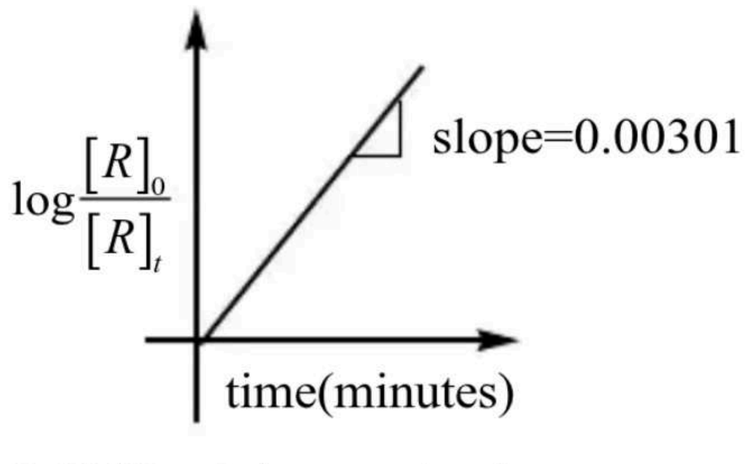
60. Van't hoff factor of phenol dissolved in certain solvent is 0.8. Degree of dimerisation of phenol is
(1) 0.8 (2) 0.4
(3) 0.2 (4) 0.5

61. 39 grams of benzene is mixed with 92 grams of Toluene to form a solution. If the vapour pressure of Benzene and toluene at 300 K are 720 mm and 450 mm respectively then mole fraction of Benzene in vapour phase will be
(1) 0.28 (2) 0.63
(3) 0.82 (4) 0.44

62. What weight of Hydrogen can be deposited by the quantity of electricity that deposits 63.6 grams of copper from $CuSO_4$ solution (Atomic mass of copper = 63.6u)
(1) 2 grams (2) 4 grams
(3) 8 grams (4) 1 grams

63. $Zn/Zn^{2+}(0.01M)||Ag^+(0.1M)/Ag$
The emf of the cell is $\left(E^0_{cell}=1.56V\right)$
(1) 1.53 V (2) 1.56 V
(3) 1.62 V (4) 1.5 V

64. For a first order reaction,



Half life of the reaction is
(1) 20 minutes (2) 10 minutes
(3) 100 minutes (4) 200 minutes

65. Which one of the following is most reactive towards electrophilic substitution ?
(1) Nitrobenzene (2) p-Xylene
(3) o-Xylene (4) m-Xylene

66. Identify incorrect statements
A) Activation energy depends nature of reactants and catalyst
B) Rate constant varies with concentration of reactants and temperature
C) Units of rate of reaction depends on order of the reaction
(D) Time taken for completion of first order reaction is ten times its half life
(1) A,B,C and D (2) C and D only
(3) A and D only (4) B,C and D only

67. Assertion (A) Dinitrogen is chemically inert
(R) : Ionization enthalpy of nitrogen is more than that of oxygen
(1) A and R are true, R explains A
(2) A and R are true, R does not explain A
(3) A is true but R is false
(4) A is false but R is true

68. Products obtained by the disproportionation of selenium monochloride are
(1) $SeCl_2, Se$ (2) $SeCl_4, SeCl_2$
(3) $SeCl_4, Se$ (4) $SeCl_2, Se_2Cl_2$

69. Regarding HF, incorrect statement is
(1) It has the highest boiling point among the hydrides of 17th group
(2) It is the strongest reducing agent among the hydrides of 17th group
(3) It can associate through hydrogen bonding even in vapour state
(4) It is thermally most stable hydride of 17th group

70. Density is highest for
(1) Cu (2) Fe
(3) Cr (4) Sc

71. Statement-I : CuF cannot exist
Statement-II : $E^0_{Zn^{2+}/Zn}$ is more negative than expected.
(1) S-I and S-II are correct
(2) S-I is incorrect but S-II is correct
(3) S-I is correct but S-II is incorrect
(4) S-I and S-II are incorrect

72.

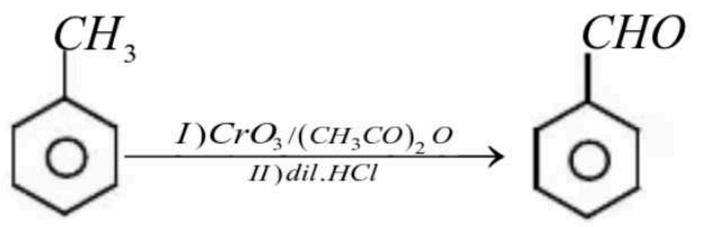
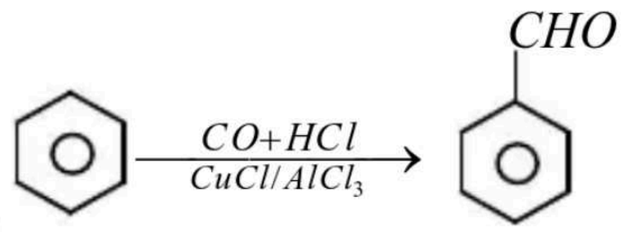
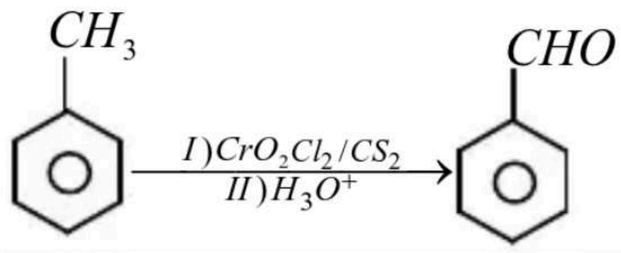
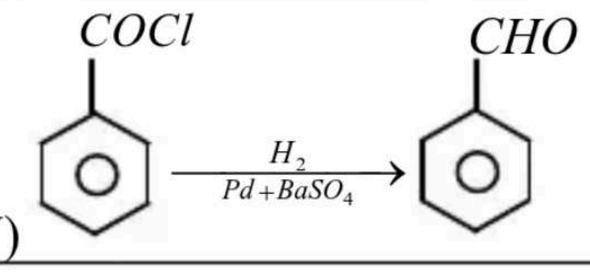
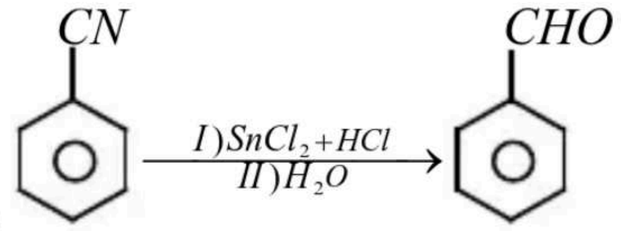
Column-I (Reaction)	Column-II (major product)
I) $R-X \xrightarrow{AgCN}$	A) R-NC
II) $R-X \xrightarrow{KNO_2}$	B) R-ONO
III) $R-X \xrightarrow{AgNO_2}$	C) $R-NO_2$
IV) $R-X \xrightarrow{KCN}$	D) R-CN

Correct match is

	I	II	III	IV
(1)	A	B	C	D
(2)	A	C	B	D
(3)	D	B	C	A
(4)	D	C	B	A

73. $CH_3COOH\,(X) \xrightarrow{LiAlH_4} A \xrightarrow[H^+]{X} B \xrightarrow{H_2O/H^+}$ final products.
Final products are
(1) 'A' and 'X'
(2) 'A' and homologue of 'X'
(3) Two moles of 'A'
(4) Two moles of 'X'

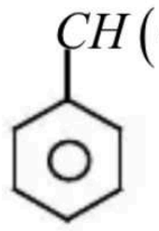
74.

Column-I (Reaction)	Column-II (Name of the reaction)
I) 	A) Gattermann-Koch
II) 	B) Rosen mund's
III) 	C) Etard
IV) 	D) Stephen's
V) 	

Correct match is

	I	II	III	IV	V
(1)	C	A	-	B	D
(2)	-	B	C	A	D
(3)	C	D	-	B	A
(4)	-	A	C	B	D

75.



$$\xrightarrow[II)dil.HCl]{I)O_2(air)} A+B;$$

'B' gives positive 2,4-DNP test.

Incorrect statement about 'A' is

(1) Liberates H_2 with sodium

(2) Forms conjugated diketone in presence of acidified sodium dichromate

(3) Forms Benzene with CH_3MgBr

(4) Forms salicyclic acid when treated with $NaOH, CO_2 / H^+$ at 6-7 atm

76. $CH_3COCl(X) \xrightarrow[II)Tollen's\ reagent/H^+]{I)H_2/Lindlar's\ catalyst} A$

$$\xrightarrow[\Delta]{P_2O_5} B \xrightarrow{H_2O} C$$

Incorrect statement is

(1) A and C are same compounds

(2) B is a good acylating agent

(3) $A \xrightarrow{SOCl_2} X$

(4) B contains two carbon atoms

77. Statement-I : Aniline cannot be prepared by Gabriel phthalimide synthesis

Statement-II : Aniline cannot be prepared by Hoffmann's hypobromite reaction.

(1) S-I and S-II are correct

(2) S-I is incorrect but S-II is correct

(3) S-I is correct but S-II is incorrect

(4) S-I and S-II are incorrect

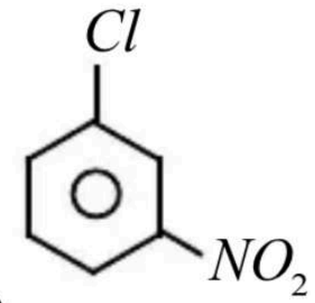
78.

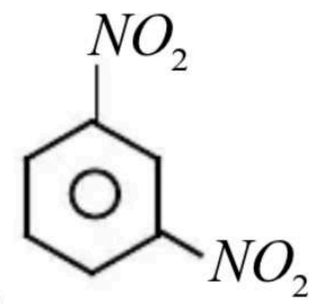


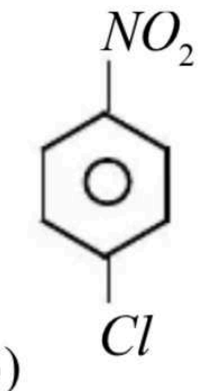
$$\xrightarrow[dark]{Cl_2/FeCl_3} A \xrightarrow{Conc.HNO_3+Conc.H_2SO_4}$$

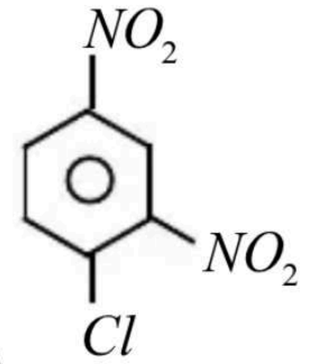
$$B(maj) \xrightarrow[II)NaNO_2+HCl/278K]{I)Sn/HCl} C \xrightarrow[II)NaNO_2+Cu/\Delta]{I)HBF_4} D$$

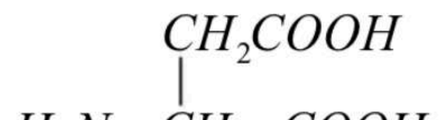
'D' is

(1) 

(2) 

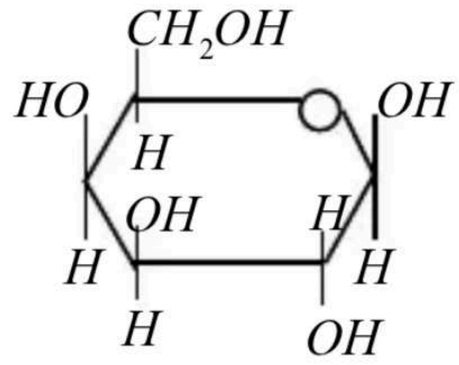
(3) 

(4) 

79. The structure  is

(1) Lysine (2) Asparatic acid

(3) Glutamine (4) Serine

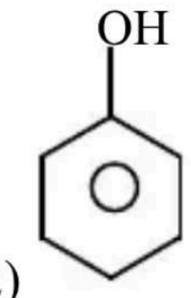
80. The structure  is

(1) $\beta-D-$galactose (2) $\beta-D-$glucose

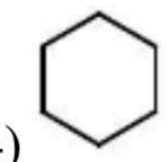
(3) $\alpha-D-$galactose (4) $\alpha-D-$glucose

81. Most acidic hydrogen is present is

(1) 

(2) 

(3) $CH_2=CH_2$

(4) 

82. Maximum number of electrons with m = 0 and $S=+\frac{1}{2}$ in chromium atom is

(1) 5 (2) 7

(3) 4 (4) 2

83. Magnitude of electron gain enthalpy 'Cl' atom is 'x' . Ionization enthalpy of $Cl^-(g)$ is

(1) $> x$

(2) x

(3) $< x$

(4) cannot be predicted

84. Hybridisation of central atom is same in all the following except.

(1) $NH_2^{\ominus}$ (2) SiO_2

(3) H_3O^+ (4) SCl_4

85. A hydrocarbon contains 81.8% by mass of carbon. The empirical formula of the hydrocarbon is

(1) C_3H_8 (2) CH_2

(3) CH_4 (4) CH

86. 0.045 M solution of urea is isotonic with 0.02 M $KHSO_4$ solution. % ionization of bisulphate ion is

(1) 25 (2) 50

(3) 75 (4) 40

87. A process is spontaneous when magnitude of $\Delta H > T\Delta S$, then the correct "signs" of ΔH and ΔS could be respectively.

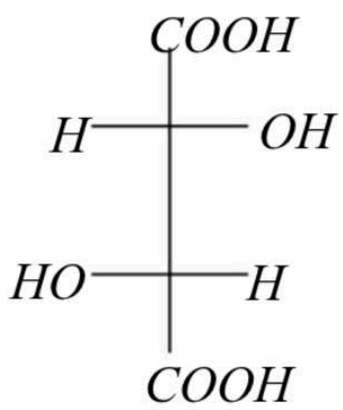
A) +, + B) –, –

C) +, – D) –, +

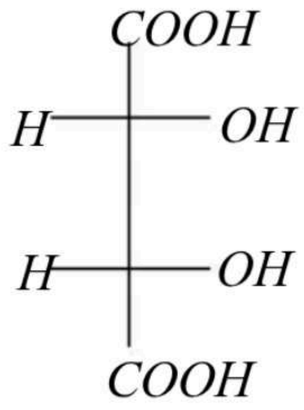
(1) A,B,C,D (2) A,C only

(3) B,D only (4) A,D only

88.



(A)



(B)

'A' and 'B' is a pair of

(1) Enantiomers

(2) Diastereomers

(3) Homomers

(4) Constitutional isomers

89.

Column-I (Homologous series)	Column-II (Characteristic reaction)
I) Carbonyl compounds	A) Nucleophilic substitution
II) Alkenes	B) Electrophilic substitution
III) Aromatic compounds	C) Electrophilic addition
IV) Alkyl halides	D) Nucleophilic addition

	I	II	III	IV
(1)	C	D	B	A
(2)	D	C	A	B
(3)	D	C	B	A
(4)	C	D	A	B

90. How many are soluble in water.

$HCHO, CH_3OH, CH_3COOH,$

$CH_3CHO, C_2H_5OH, CH_3COCH_3$

(1) 6 (2) 2

(3) 4 (4) 3

BIOLOGY

91. The type of muscle fibres which are present in the wall of stomach and Intestine are characterized by the presence of

a) Cylindrical and elongated shape

b) Communication junctions

c) Centrally placed nucleus

d) Syncytial condition

Select the option with the correct set

(1) a, b, c and d (2) a and b only

(3) a and d only (4) b and c only

92. Family of cat is

(1) Felidae (2) Cannidae

(3) Hominidae (4) Muscidae

93. Neurospora, Claviceps and Truffles belong to
(1) Phycomycetes (2) Deuteromycetes
(3) Ascomycetes (4) Basidiomycetes

94. Assertion (A) : Visceral organs located in the haemocoel are bathed in haemolymph in *Periplaneta.*

Reason (R) : Cockroach has an open circulatory system and blood in haemocoel transports the oxygen to different parts of body.

(1) Both A and R are true, and R is the correct explanation of A

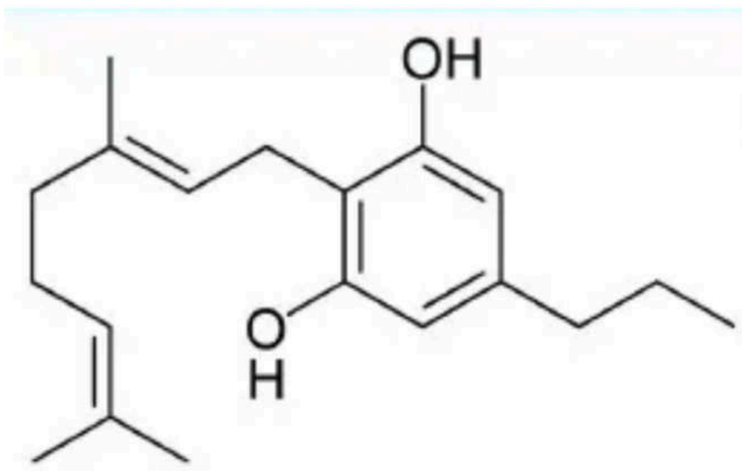
(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

95. Haploid spores of pteridophytes give rise to
(1) Free-living gametophyte
(2) Diploid gametophyte
(3) Free-living sporophyte
(4) Sporophyte attached to the parent plant

96. Select the incorrect option w.r.t. the given structure



(1) Naturally obtained from the inflorescences of the plant *Cannabis sativa*.

(2) Known for their effects on cardio – vascular system our body.

(3) Inhibits the transport of the neuro – transmitter dopamine.

(4) Generally taken by inhalation and oral ingestion

97. Length of chloroplast is approximately
(1) 2-4 μm (2) 5-10 μm
(3) 20-25 μm (4) 100 μm

98. In a population of rabbit it is observed that average natality is 150, average mortality is 100, immigration is 50 and emigration is 20. What will be the net population after time t, if initial population had 200 individuals?
(1) 520 (2) 1600
(3) 280 (4) 680

99. Match the following columns

Column – I		Column – II	
A.	Adaptive radiation	I.	Selection of resistant varieties due to excessive use of herbicides and pesticides.
B.	Convergent evolution	II.	Bones of forelimbs in man and whale
C.	Divergent evolution	III.	Wings of butterfly and of bird
D.	Evolution by anthropogenic action	IV.	Darwin finches

Choose the correct answer from the options given below.
(1) A – I, B – IV, C – III, D – II
(2) A – IV, B – III, C – II, D – I
(3) A – III, B – II, C – I, D – IV
(4) A – II, B –I, C – IV, D – III

100. Which of the following cellular parts is correctly described ?

Column-I	Column-II
a) SER	i) Basal body of Cilia
b) Golgi apparatus	ii) Protein synthesis
c) Ribosome	iii) Steroidal hormone synthesis
d) Centrosome	iv) Glycolipid synthesis

(1) a(ii), b(iv), c(iii), d(i)
(2) a(ii), b(iii), c(iv), d(i)
(3) a(iv), b(ii), c(i), d(iii)
(4) a(iii), b(iv), c(ii), d(i)

101. Select the correct statement w.r.t the life cycle of plasmodium.

(1) Sporozoites first enter into spleen before starting divisions in humans

(2) Asexual reproduction occurs only in liver cells of human host.

(3) Gametocyte stage is formed in the salivary glands of mosquito

(4) Fertilisation and development take place in gut of mosquito

102. Which of the following features indicate(s) sexual dimorphism in Frogs?

a) Presence of Bidder's canal in males

b) Presence of vocal sacs in males

c) Presence of copulatory pads on the first digit of the forelimbs in males

Choose the correct option

(1) Only a and b (2) Only b and c

(3) Only a and c (4) a, b, c

103. Among the following characteristics, how many are correct for red algae ?

a. Has phycoerythrin

b. Mostly fresh water algae

c. Chloroplast without pyrenoid

d. Sexual reproduction oogamous type

(1) Four (2) Three

(3) Two (4) One

104. An old man was hospitalised due to difficulty in breathing. He had symptoms such as congestion of the lungs and reduced cardiac output. He is most likely suffering from the disorder in which heart is unable to pump blood effectively known as

(1) Emphysema (2) Heart failure

(3) Heart attack (4) Angina

105. Assertion (A) : In grasses bulliform cells are present on adaxial surface along the veins.

Reason (R) : When bulliform cells are turgid, leaf surface is exposed and when they are flaccid, they make the leaf curl inwards to minimise water loss.

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

106. Assertion (A) : During the process of fertilization only one sperm can enter ovum (secondary oocyte) under normal conditions.

Reason (R) : Sperm after binding with its receptor present on zona pellucida layer of the ovum (secondary oocyte) induces changes in the membrane that blocks the entry of additional sperms.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

107. Fertilization of gametes does not occur in petri dish / test tube in a technique called

(1) Zygote – intra fallopian transfer

(2) Intra – uterine transfer

(3) Gamete – intra fallopian transfer

(4) Intra cytoplasmic sperm injection

108. Select the incorrect match.

Column-I	Column-II
a) Abrin	i) Concanavalin A
b) Receptor	ii) Toxin
c) GLUT-4	iii) Hormone
d) Lectins	iv) Transport protein

(1) a(ii), b(i), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(iii), b(ii), c(i), d(iv)

109. Calculate the number of chromosomes if a cell has 20 bivalents in prophase I.

(1) 40 (2) 80

(3) 20 (4) 60

110. Which of the following is an incorrect match with respect to the animals, level of organisation, circulatory system and respiratory system?

	Animals	Level of organisation	Circulatory system	Respiratory system
(1)	Scorpion	Organ system	Open	Book lungs
(2)	Prawn	Organ system	Closed	Gills
(3)	Fasciola	Organ level	Absent	Absent
(4)	Locusta	Organ system	Open	Tracheal tubes

111. Statement I: Some cells in adult animals exit G1 phase and enter quiescent stage, but in higher plants, no such stage of any cells is seen.

Statement II: The cell in an organism that enters an inactive phase, called quiescent stage, is also an arrested part of cell cycle.

In the light of above statements, choose the correct answer from the option given below:

(1) Both, statement I and statement II are true

(2) Both, statement I and statement II are false

(3) Statement I is true and statement II is false

(4) Statement I is false and statement Il is true

112. Polysaccharides are long chain of

(1) Amino acids (2) Nucleotides

(3) Fatty acids (4) Sugars

113. Identify the examples which is not true w.r.t commensalism

a) Mycorrhizae

b) Sea – anemone and clown fish

c) Barnacle and whale

d) Fig tree and wasp

e) Goat and Abingdon tortoise

Choose the correct option from the following

(1) b, c and d (2) a, d and e

(3) a, c and e (4) All except "e"

114. Given below are two statements:

Statement – I : During summer when body loses a lot of water leading to dehydration, the release of ADH is suppressed.

Statement – II : When someone drinks a lot of water release of vasopressin is suppressed.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Only statement I – is correct, statement – II is incorrect

(4) Only statement I – is incorrect, statement – II is correct

115. Statement I : An enzyme like any protein has the secondary and tertiary structure.

Statement II : Competitive inhibitors of enzymes are often used in the control of viral pathogens.

(1) Only statement I is correct

(2) Only statement II is correct

(3) Both statement I and II are correct

(4) Both statement I and II are incorrect

116. Which of the following happens during muscle contraction?

(1) Length of "A" band increases

(2) Length of "I" band decreases

(3) Length of "A" band decreases

(4) Length of "H" band increases

117. The following are the statements about chordates.

A) Pharynx is perforated by gill slits

B) Notochord is present

C) Central nervous system is ventral and solid

D) Heart is ventral

E) Post anal tail is absent

Choose the correct statements from the options given below.

(1) A and C only (2) A, B and D only

(3) B, D and E only (4) B, C and D only

118. The $\overset{\frown}{C_{(5)}\ A_5}$ condition is found in the family
(1) Liliaceae (2) Poaceae
(3) Solanaceae (4) Fabaceae

119. Select the incorrect statement about Stamens in angiosperms
(1) Monoadelphous stamens are found in Hibiscus
(2) Pea plant has diadelphous stamens
(3) Mustard stamens show variation in length of filaments
(4) A sterile pistil is called pistillode

120. Match the following columns

Column – I		Column – II	
A.	ELISA	I.	Amplification of minute quantities of DNA
B.	PCR	II.	Identification of mutated genes after usage of probe
C.	Auto – radiography	III.	Modern method of diagnosis that serves the purpose of early diagnosis
D.	Serum analysis	IV.	Conventional method of diagnosis

Choose the correct answer from the options given below.
(1) A – IV, B – III, C – II, D – I
(2) A – III, B – I, C – II, D – IV
(3) A – IV, B – I, C – II, D – III
(4) A – II, B –IV, C – III, D – I

121. 2,4 –Dichlorophenoxyacetic acid is a
(1) Synthetic auxin
(2) Natural plant hormone
(3) Gaseous hormone
(4) Carotenoid derivative

122. Consider the given parts of a kidney.
a) PCT
b) DCT
c) Thin descending limb of Henle's loop
d) Collecting duct.
Under normal physiological conditions, the selective secretion of H^+ occurs mainly in
(1) b and c oly (2) b and d only
(3) a, b and d only (4) a, b and c only

123. Epipetalous stamens are found in
(1) Mustard (2) Pisum
(3) Tulip (4) Brinjal

124. In humans, Total lung capacity (TLC) represented by
(1) ERV + RV (2) FRC + IC
(3) VC – RV (4) EC + RV

125. Which of the following is incorrect statement about the non – iodised hormone secreted by thyroid gland.
(1) It is secreted by para follicular cells
(2) It is secreted in response to hyper calcemia
(3) It is a peptide hormone
(4) It increases blood calcium levels

126. Given below are two statements:
Statement – I : Rh incompatibility between Rh^{-ve} mother and Rh^{+ve} foetus can lead to erythro - blastosis foetalis in baby.
Statement – II : AB^{+ve} person can accept blood from $AB^{-ve}, A^{+ve}, B^{-ve}$ and O^{+ve} person.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Only statement I – is correct, statement – II is incorrect
(4) Only statement I – is incorrect, statement – II is correct

127. A pigment appears yellow-orange in the chromatogram and found in leaves of higher plants, helps
a. In absorption of light for photosynthesis
b. To prevent oxidative destruction of chlorophyll pigment by light
c. By absorbing light wavelengths beyond visible spectrum which increases the efficiency of photosynthesis
(1) only a (2) only a & b
(3) all a, b & c (4) only b & c

128. Number of embryosac in each ovule is usually

(1) 1 (2) 7

(3) 8 (4) More than 1

129. Find the incorrect match w.r.t. sacred groves and their locations.

(1) Khasi and Jaintia hills – Meghalaya

(2) Aravali hills – Rajasthan

(3) Western Ghats – Karnataka

(4) Bastar – Maharashtra

130. The most dramatic period of the cell cycle is

(1) G_1 Phase (2) S Phase

(3) G_2 Phase (4) M Phase

131. Study the following statements and choose the correct one.

a) The process of child birth is called parturition which is induced by a complex neuro – endocrine mechanism involving cortisol, estrogen and prolactin.

b) Mammary glands differentiate during pregnancy and secrete milk after child birth.

c) During embryonic development, the movement of blastomeres is most prominently associated with the morula stage.

d) Oviducts, uterus and vagina are female accessory ducts.

(1) 'a' only (2) b and d only

(3) b and c only (4) a, b and d only

132. Four genes a, b, c and d are present on a chromosome. Percentage of recombination between a and b = 15%; a and d = 2%, c and b = 8% and d and b = 13%. What will be the sequence of genes on the chromosome?

(1) adbc (2) cdab

(3) dcab (4) acbd

133. Match the following columns

Column – I		Column – II	
A.	Down's syndrome	I.	Sterile with overall masculine development
B.	Turner's syndrome	II.	Short stature with a small round head
C.	Klinefelter's syndrome	III.	Ovaries are rudimentary
D.	Phenyl ketonuria	IV.	Reduction in hair and skin pigmentation

Choose the correct answer from the options given below.

(1) A – II, B – I, C – III, D – IV

(2) A – II, B – III, C – IV, D – I

(3) A – II, B – III, C – I, D – IV

(4) A – I, B –IV, C – III, D – II

134. Select the correct match w.r.t parts of brain and their function.

(1) Cerebellum – regulates body temperature

(2) Thalamus – major co-ordinating centre for sensory and motor signalling.

(3) Hypothalamus – respiratory rhytum centre

(4) Medulla oblongata – Memory and communication

135. How many statements are correct :

(a) G_2 phase is the period of cytoplasmic growth

(b) Animals never show equational division in haploid cells

(c) Meiosis occurs in diploid cells which are destined to form gametes

(d) Homologous chromosomes remain linked always even after the end of crossing over

(e) Identity of chromosome is lost as discrete element during telophase

(1) 2 (2) 3

(3) 4 (4) 5

136. Primary productivity of an area depends on

a) Photo – synthetic capacity of plants

b) Plant species inhabiting that area

c) Availability of nutrients

d) Variety of environmental factors.

The correct ones are

(1) Only a and b

(2) Only c and d

(3) Only b, c and d

(4) All a, b, c and d

137. Assertion (A):- Most of the enzymes are functional in tertiary structure

Reason (R):-In tertiary structure the backbone of protein chain folds upon itself, the chain cris-crosses itself and hence many crevices are made and one such crevice is called active site into which the substrate fits.

. (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

138. In a transcription unit, the promoter is located towards

(1) 5' end of the templates strand

(2) 5' end of the coding strand

(3) 3' end of the non-template strand

(4) 3' end of the sense strand

139. In case of ADA deficiency, which technique is considered completely curative?

(1) Gene theraphy in early embryonic stage

(2) Bone marrow transplantation

(3) Enzyme replacement therapy

(4) Treatment with antibiotics

140. The RNA polymerase III transcribes

(1) tRNA (2) 28S rRNA

(3) 5.8S rRNA (4) Pre- mRNA

141. Assertion (A): If an inheritable mutation is observed in a population at high frequency it is referred to as DNA polymorphism.

Reason (R): Allelic sequence variation is described as a DNA polymorphism if more than one variant at a locus occurs in human population with a frequency less than 0.01

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

142. The trait which is generally controlled by two or more genes, inheritance of such traits is called

(1) Multiple allelism

(2) Pleiotropy

(3) Polygenic inheritance

(4) Epistasis

143. Select the option that correctly identifies A, B and C in the given table.

Organism	Trophic level	Food chain
Eagle	A	Grazing
Earthworm	Primary consumer	B
Frog	C	Grazing

(1) A – Top carnivore, B – Detritus, C – Secondary consumer.

(2) A – Top carnivore, B – Detritus, C – Primary consumer.

(3) A – Secondary consumer, B – Grazing, C – Secondary consumer.

(4) A – Scavenger, B – Grazing, C – Producer

144. Open circulatory system present in

(1) Hirudinaria and Octopus

(2) Limulus and Pila

(3) Corvus and Aptenodytes

(4) Chelone and Nereis

145. Number of ATP synthesized in one turn of Kreb`s cycle

(1) 1 (2) 2

(3) 4 (4) 6

146. Choose the odd one w.r.t. auto – immune disorders.

(1) Psoriasis

(2) Rheumatoid arthritis

(3) Alzheimer's disease

(4) Vitiligo

147. Characteristics of a flower are as

a. Nectar present

b. Flower are fragrant.

c. Flowers are large and colourful

d. Sticky pollen grains

On the basis of above features, flower must be

(1) Hydrophilous (2) Anemophilous

(3) Entomophilous (4) Cleistogamous

148. Match List I with List – II and choose the correct option

List – I		List – II	
A.	Androgens	I.	Suppress the immune responses
B.	Insulin	II.	High pitch of voice in females
C.	Estrogen	III.	Anabolic effects on protein and carbohydrate metabolism in males
D.	Cortisol	IV.	Enhances cellular glucose utilisation

Choose the correct answer from the options given below

(1) A – II, B – I, C – IV, D – III

(2) A – III, B – IV, C – II, D – I

(3) A – II, B – I, C – III, D – IV

(4) A – IV, B – III, C – II, D – I

149. During cytokinesis, a furrow in the plasma membrane develops and divides the cell cytoplasm into two in

(1) Chlamydomonas

(2) Human

(3) Generative cell of pollen

(4) Cladophora

150. Genes in eukaryotes

(a) Have introns

(b) Are split

(c) Are mostly polycistronic

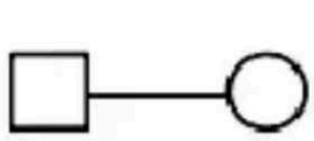
(1) (a) & (b) only

(2) (b) & (c) only

(3) (a) & (c) only

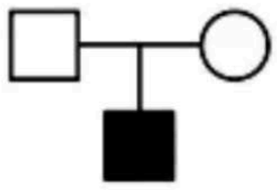
(4) All (a), (b) & (c)

151. Which one of the following symbols represents matching between relatives in human pedigree analysis?

(1) 

(2) 

(3) 

(4) 

152. Given below are two statements:

Statement – I : Lactational amenorrhoea has been reported to be effective upto a period of six months following parturition.

Statement – II : Side effects are almost nil and failure rate is high in case of natural methods of contraception.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Only statement I – is correct, statement – II is incorrect

(4) Only statement I – is incorrect, statement – II is correct

153. Genetic code is degenerate, which means
(1) One codon codes for only one amino acid
(2) Codon is read on mRNA in a contiguous fashion
(3) Three codons do not code for any amino acid
(4) Some amino acids are coded by more than one codon

154. Read the following statements and choose the correct option
Statement-A : Not all characters show true dominance
Statement-B : ABO blood group shows dominance, incomplete dominance and multiple allelism.
(1) Only A is correct
(2) Only A is incorrect
(3) Both A and B are incorrect
(4) Both A and B are correct

155. Choose the incorrect option w.r.t the total number of respective bones in an adult human?
(1) Unpaired bones of cranium – 4
(2) Vertebro – chondral ribs + floating ribs – 10
(3) Facial bones + cervical vertebrae – 21
(4) Carpals + meta carpals – 18

156. Find the incorrect match w.r.t periplaneta americana
(1) No.of spiracles – 10 pairs
(2) No of moultings – 13
(3) Brood pouch in females – formed by 7th, 8th, 9th terga
(4) No of chambers in heart – 13

157. Match the column I with column II and choose the correct option.

Column-I	Column-II
a) Totipotency	i) Pomato
b) Protoplast fusion	ii) Unorganised mass of cell
c) Explant	iii) Capacity to generate whole plant from any living cell
d) Callus	iv) Plant part which is cultured

(1) a(i), b(ii), c(iii), d(iv)
(2) a(iii), b(i), c(iv), d(ii)
(3) a(iii), b(iv), c(i), d(ii)
(4) a(ii), b(i), c(iv), d(iii)

158. Refer to the following information about the human female reproductive system.
Each primary ocyte gets surrounded by a layer of granulosa cells to form the primary follicles, these primary follicles later develop additional layers of granulosa cells and a theca to become secondary follicles.
Choose the correct answer from the following
(1) The statement is true for primary follicle formation but false for secondary follicle development
(2) The statement is true for both primary and secondary follicle formation
(3) The statement is false for primary follicle formation but true for secondary follicle development
(4) The statement is false for both primary and secondary follicle formation

159. Assertion(A) : Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis
Reason(R) : Nucleolus is the active site of ribosomal RNA synthesis
(1) Both (A) & (R) are correct but (R) is the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is not the correct explanation of (A).

160. Contraction of diaphragm during inspiration leads to

(1) Increase in the volume of thoracic – chamber in the dorso – ventral axis

(2) Increase in the volume of thoracic chamber as well as pulmonary cavity in the antero – posterior axis.

(3) Decrease in the volume of pulmonary cavity but increase in that of thoracic chamber.

(4) No change in the volume of thoracic chamber

161. Match the following columns

Column – I		Column – II	
A.	Habitat loss and fragmentation	I.	Occurs in organisms associated in an obligatory way
B.	Co – extinction	II.	Cause of extinction of Stellar's sea cow
C.	Allien species invasion	III.	*Eichornia* in India
D.	Over – exploitation	IV.	Most important cause of bio – diversity loss

(1) A – IV, B – II, C – I, D – III

(2) A – IV, B –I, C – III, D – II

(3) A – II, B – IV, C – I, D – III

(4) A – II, B –I, C – IV, D – III

162. Presence of all of the following factors are known to affect Hardy – Weinberg equilibrium, except

(1) Genetic recombination

(2) Gene flow

(3) Natural selection

(4) Random mating

163. Match the following

Column-I	Column-II
A) Primase	i) Enzymatic synthesis of RNA
B) Permease	ii) Peptide bond formation
C) Polynucleotide phosphorylase	iii) Primer formation in DNA replication
D) Ribozyme	iv) Increases permeability of cell to Beta-galactosides

(1) A(iii), B(iv), C(i), D(ii)

(2) A(iv), B(iii), C(ii), D(i)

(3) A(iv), B(iii), C(i), D(ii)

(4) A(ii), B(i), C(iii), D(iv)

164. Read the following statements.

Statement A: Today, we know more than 900 restriction enzymes isolated from over 20 strains of eukaryotes, each of which recognises the same recognition sequence.

Statement B: If a 3 kb dsDNA undergoes 30 cycles of PCR, then 1 billion copies of this dsDNA will be obtained.

Choose the correct option.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

165. The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid of

(1) *Salmonella typhimurium*

(2) *Escherichia coli*

(3) *Agrobacterium tumefaciens*

(4) *Bacillus thuringiensis*

166. Sauropsid vertebrates shows

(1) Internal fertilization and internal development

(2) Internal fertilization and external development

(3) External fertilization and internal development

(4) External fertilization and external development

167. Assertion (A): When we describe the story of life on earth, we treat evolution as a consequence of a process called natural selection.

Reason (R): Natural selection is not at all based on factual observations and is independent of availability of natural resources (or) seasonal fluctuations.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

168. Malignant tumors

(1) Are less invasive than benign tumors

(2) Exhibit contact inhibition

(3) Remain confined to their original location

(4) Exhibit metastasis

169. Select the option having correct set of redifferentiated tissues

(1) Wound cambium, interfascicular cambium

(2) Phellogen, Vascular cambium

(3) Parenchyma, collenchyma, sclerenchyma

(4) Secondary xylem, Secondary phloem

170. A man with normal vision marries a woman who is not colourblind but has a colourblind father. What is the probability that their children will be colour blind?

(1) 50% sons only

(2) 25 % daughter only

(3) 50% daughters only

(4) 50% of sons and 50% of daughters

171. Read the following statements w.r.t. restriction endonucleases:

(a) Each of them functions by inspecting the length of a DNA sequence.

(b) Each of them recognises a specific sequence of base pairs in DNA that reads the same on the two strands when the orientation of reading is kept the same.

(c) They always cut the strands of DNA far away from the centre of palindromic sites.

(d) These enzymes specifically cut the hydrogen bonds to get the desired DNA fragments.

How many of the above given statements is/are incorrect w.r.t. them?

(1) One (2) Two

(3) Three (4) Four

172. A 25 years old female goes to doctor. She asks doctor that after 4 days there is due date of her menses and she want to postpone bleeding for some days as she will be engaged in some official work during those days. Which hormone will the doctor advise her to take?

(1) FSH (2) Oxytocin

(3) Progesterone (4) Relaxin

173. In lac operon, when repressor protein binds to operator then

(1) There is synthesis of β-galactosidase enzyme

(2) RNA polymerase cannot transcribe operon

(3) RNA polymerase binds to the promoter and starts transcription

(4) Only `y' gene is translated

174. The region of Biosphere reserve which is legally protected and where no human activity is allowed is known as

(1) Buffer zone

(2) Transition zone

(3) Restoration zone

(4) Core zone

175. A sequence of nucleotides from where replication starts and any piece of DNA when linked to this, can be made to replicate within the host cell, is known as

(1) Selectable marker

(2) Recognition site

(3) Cloning site

(4) Origin of replication

176. A biochemist isolates and purifies various molecules needed for DNA replication. When he adds some DNA, replication occurs, but each DNA molecule consists of a normal strand paired with numerous segments of DNA a few hundred nucleotides long. What has he probably left out of the mixture?

(1) DNA polymerase

(2) DNA ligase

(3) nucleotides

(4) Okazaki fragments

177. Assertion (A): The effect of water as a factor is more through its effect on the plant, rather than directly on photosynthesis.

Reason (R): Very less amount of absorbed water is used in photosynthesis and any water stress causes the stomata to close hence reducing the CO_2 availability and also makes the leaves wilt.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

178. A. The respiratory quotient depends upon the type of respiratory substrate used during respiration.

B. When carbohydrates are used as substrate and are completely oxidised, the RQ will be 1.

C. Pure fats can be used as respiratory substrate but pure protein can`t.

D. First cell of the earth planet lived in aerobic condition

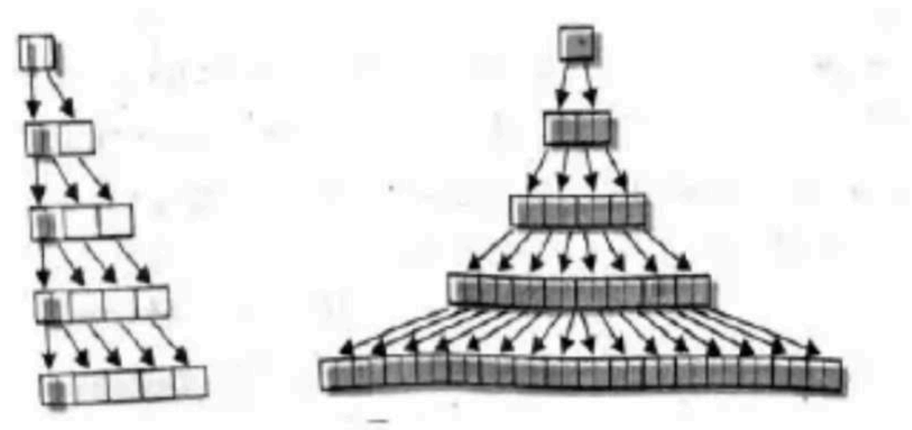
E. Inner mitochondrial membrane has no enzyme for Kreb`s cycle but has for ETS

In the above statements , how many are incorrect ?

(1) 1 (2) 2

(3) 3 (4) 5

179. Identify the growth type in the given diagram:



A B

(1) A-Geometric growth, B–Arithmetic growth

(2) A-Arithmetic growth, B-Geometric growth

(3) A-Logistic growth, B-Geometric growth

(4) A- Sigmoid growth, B-Linear growth

180. Which one of the following statements is wrong for the plants showing 'Kranz anatomy'?

(1) Bundle sheath cells are present

(2) The cells surrounding the vascular bundle have a large number of chloroplasts

(3) Primary CO_2 acceptor is a C_4-acid

(4) RuBisCO is present in stroma of chloroplast of bundle sheath cell